$neAL$

23 **B** When the capacitor is fully charged,
i) the p.d. across it is the same as the p.d. across the battery or across the 4.0 Ω resistor;
ii) no current flows through it and hence ammeter A_2 reading = 0 A.

Current only flows in the upper loop.

Ammeter A_1 reading $= \dfrac{4.0}{(4.0+1.0)} = 0.80$ A

Terminal p.d. $= 4.0 - (0.80 \times 1.0) = 3.2$ V

24 **C** $BIL\sin\theta - mg = ma$